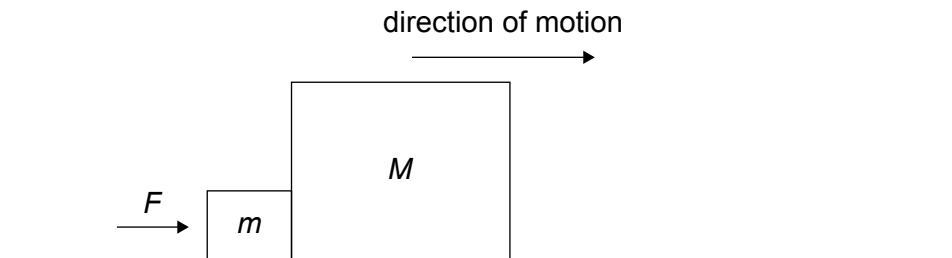


PAPER 1 / QUESTION 42

25 October 2023 / TZ2 (file label) / HL Paper 1 / original question 4 / 8823-6513

42. The diagram shows two blocks of mass m and M in contact on a frictionless surface. A force F is applied on the block of mass m and causes the blocks to move with an acceleration a .



What is the force that the block of mass M is exerting on the block of mass m ?

- A. ma
- B. Ma
- C. $(M + m)a$
- D. $\frac{m}{M + m}a$